

AKSHAY PRALHAD KANADE

Address: Prüfeninger Strasse 61, 93049 Regensburg, Germany
Mobile: (+49) 15215423232
E-Mail: akshaykanade77@gmail.com
LinkedIn: www.linkedin.com/in/akshay-kanade-746398122
Portfolio: akshay-kanade.com



PROFESSIONAL EXPERIENCE

Independent AI/ML Engineer - Projects & Technical Development - Regensburg, Germany *Nov 2025 - Present*

- Built an AI-powered Product Recognition Dashboard using React, TypeScript, n8n, and Groq LLM, combining free-form product descriptions, fuzzy matching, and conversational AI.
- Automated 5+ manual steps in the job application process using AI agents and n8n, including company research, recruiter selection, language detection, personalized email generation, CV selection, and application tracking.
- Developed AI Incident Commander, an autonomous LLM agent with dynamic tool execution, human approval workflows, and memory-based learning for automated incident response.
- Built and experimented with 2+ GenAI/RAG applications using embeddings, vector search, LLMs, and FastAPI, while continuing professional development through German B1 studies and AI/ML certifications.

Software Developer (AI/ML Engineer) - OneVision Software AG, Regensburg *Jan 2024 - Nov 2025*

- Designed end-to-end perception pipelines from ingestion to deployment, shortening delivery cycles by 50%.
- Boosted inference throughput by 70% through TensorRT tuning, ONNX export, and mixed precision execution.
- Benchmarked CNN, ViT, and foundation models, raising model selection efficiency by 40% across industrial imaging tasks.
- Introduced explainability and validation layers, reducing model related support queries by 30% and strengthening stakeholder trust.
- Coordinated with platform, DevOps, and product teams to operationalize models via CI/CD, enabling weekly production rollouts.

Working Student & Master Thesis (ADAS Validation, Computer Vision & ML Engineering)

- AVL Software and Functions GmbH, Regensburg *Oct 2022 - Dec 2023*

- Built containerized object detection pipelines for ARM platforms, cutting inference latency by 50%.
- Reduced model footprint by 40-60% using pruning and quantization, enabling real time edge deployment.
- Automated ROS bag processing and annotation flows with Python/SQL, saving 30+ hours per month in manual effort.
- Verified LiDAR and radar outputs to ensure perception accuracy for ADAS validation across diverse test scenarios.

Data analyst - Accenture, Mumbai, India *Jan 2019 - Aug 2021*

- Created ETL workflows and automated reporting systems, improving data processing efficiency by 65%.
- Constructed SQL driven data pipelines supporting analytics and ML initiatives for enterprise clients.
- Partnered with cross functional teams to translate raw data into actionable insights for large scale operations.

EDUCATION

Hochschule Ravensburg Weingarten *Sep 2021- Oct 2023*

Master of Engineering: Electrical Engineering and Embedded Systems

Savitribai Phule Pune University *May 2014 – June 2018*

Bachelor of Engineering: Electronics and Telecommunication

SKILLS

- **Programming Languages:** Python, C++, C, SQL, JavaScript, TypeScript, HTML5, CSS3, JSON
- **AI & Machine Learning:** Supervised Learning, Unsupervised Learning, Deep Learning, Transfer Learning, Representation Learning, Model Fine-Tuning, Classification, Regression, Feature Engineering, Model Evaluation, Model Validation, CNNs, Transformers, Vision Transformers (ViTs), Foundation Models
- **Computer Vision:** Object Detection, Image Segmentation, Image Classification, Image Processing, OpenCV, YOLO, Faster R-CNN, U-Net, Detectron2, Albumentations, Camera Calibration, Dataset Curation, Data Augmentation
- **Generative AI & LLM Engineering:** Large Language Models (LLMs), Hugging Face Transformers, Prompt Engineering, Structured LLM Outputs, RAG, Embeddings, Semantic Search, Vector Databases, FAISS, Conversational AI, LLM Evaluation, Groq API
- **AI Agents & Automation:** AI Agents, Agentic Workflows, n8n, Workflow Automation, Function Calling, Structured Data Extraction, LLM-powered Applications, REST API Integration, JSON-based Workflows
- **Web & Application Development:** React.js, Next.js, Vite, Tailwind CSS, Responsive Web Applications, REST API Development
- **ML Engineering & MLOps:** End-to-End ML Pipelines, Experiment Tracking, Model Versioning, Dataset Versioning, Model Deployment, Model Monitoring, Model Drift Detection, MLflow, DVC, Docker, Kubernetes, Jenkins, CI/CD, FastAPI
- **Model Optimization & Edge AI:** ONNX, TensorRT, Pruning, Quantization, Mixed Precision, Inference Optimization, Model Compression, CPU/GPU Optimization, NVIDIA Jetson, ARM-Based Deployment, Real-Time Inference, Edge AI
- **Cloud & Infrastructure:** AWS (S3, EC2, SageMaker), Azure ML, Google Cloud Platform, Linux, Windows, Cloud-Based ML Deployment, Edge-to-Cloud Architecture
- **Data Engineering & Analytics:** ETL Pipelines, Data Cleaning, Data Transformation, Data Validation, Data Quality, Exploratory Data Analysis (EDA), Statistical Analysis, Pandas, NumPy, Scikit-learn
- **Autonomous Systems & Sensor Data:** ADAS, LiDAR Processing, Radar Data Processing, Sensor Validation, Sensor Fusion, ROS, Vehicle Perception, Object Tracking, Time-Series Analysis, Embedded AI
- **Data Annotation & Dataset Engineering:** CVAT, Label Studio, Data Annotation, Annotation Quality Control, Dataset Validation, Image Preprocessing
- **Data Visualization & Business Intelligence:** Power BI, Tableau, Matplotlib, Seaborn, Interactive Dashboards, KPI Reporting, Automated Reporting, Data Storytelling
- **Developer & Productivity Tools:** Git, GitHub, VS Code, PyCharm, Jupyter, PowerShell, Microsoft 365, Postman, Docker Desktop
- **AI-Assisted Development:** GitHub Copilot, ChatGPT, Claude, Google Gemini, AI-assisted Coding, Code Generation, Debugging, Code Review, Technical Research
- **Languages:** English (C1), German (A2 - currently learning B1)

- **AI/ML Model Optimization & Edge Deployment**

Technologies: Python, PyTorch, ONNX, TensorRT, Docker, AWS, Raspberry Pi, Pruning, Quantization

Optimized deep learning inference pipelines for cloud and edge deployment by converting PyTorch models to ONNX and applying pruning, quantization, mixed precision, and TensorRT optimization.

Achieved up to 5× faster model execution while reducing model size and computational requirements for resource-constrained edge devices.

Designed reproducible cloud-to-edge deployment workflows using Docker and CI/CD, reducing manual deployment effort by approximately 50%.

- **AI-Based Object Detection & Segmentation Pipeline**

Technologies: PyTorch, YOLOv5, Faster R-CNN, U-Net, OpenCV, Albumentations, TensorRT, Jetson Nano

Developed an end-to-end computer vision pipeline covering preprocessing, augmentation, model training, evaluation, optimization, and edge deployment across 5+ datasets.

Implemented YOLOv5, Faster R-CNN, and U-Net models, achieving >92% detection accuracy on evaluated datasets.

Optimized Jetson Nano inference using pruning and TensorRT, improving edge inference performance by approximately 60%.

- **AI-Powered Product Recognition & Agent Workflow Platform**

Technologies: React, TypeScript, n8n, Groq LLM, REST APIs, Google Sheets

Designed and implemented an AI-powered product recognition workflow using n8n, combining rule-based decision logic with Groq LLM reasoning to automate product identification and context-aware interactions. Developed workflows that transform unstructured user requests into structured JSON, enabling reliable AI decision-making, workflow orchestration, and REST API integration.

Built multilingual conversational AI with automated chat-history management, demonstrating practical experience in LLM integration, AI agents, and workflow automation.

- **LLM-Based RAG Assistant**

Technologies: Hugging Face Transformers, Embeddings, FAISS, FastAPI, Docker, Python

Developed a Retrieval-Augmented Generation system enabling natural-language queries over a knowledge base using semantic embeddings and FAISS vector search.

Designed document ingestion, chunking, retrieval, and prompt-generation workflows, improving retrieval relevance by approximately 20–30% through experimentation.

Deployed the application as a Dockerized FastAPI service for scalable, reproducible LLM inference.

- **MLOps Deployment & Monitoring Platform**

Technologies: MLflow, DVC, Docker, Git, Jenkins, Python, CI/CD

Designed an end-to-end MLOps workflow integrating experiment tracking, dataset and model versioning, containerization, automated testing, and deployment.

Used MLflow and DVC across 20+ experiments and implemented Jenkins-based CI/CD pipelines, reducing model release time by approximately 35–40%.

Built monitoring workflows for model performance, latency, and model drift to improve reliability and reproducibility across environments.

- **Vehicle Position & Dimension Prediction**

Technologies: Python, LiDAR, Velodyne, Pandas, NumPy, Scikit-learn, Time-Series ML

Developed a machine learning pipeline for estimating vehicle position and dimensions using LiDAR/Velodyne

sensor data across different driving scenarios.

Performed sensor-data preprocessing, feature engineering, dataset preparation, and regression/time-series analysis to model vehicle movement and characteristics.

Evaluated predictions across multiple scenarios, demonstrating practical experience in automotive AI, sensor data processing, and ADAS applications.

- **Camera Calibration & Image Processing Automation**

Technologies: Python, OpenCV, NumPy

Developed an automated camera calibration and image-processing pipeline for computer vision applications using OpenCV.

Performed intrinsic calibration with a reprojection error of 0.0156 and implemented edge detection, perspective correction, and geometric transformations.

Automated repetitive image-processing tasks, reducing manual image adjustment effort by approximately 50% and validating calibration across multiple camera configurations.

- **Predictive Analytics Data Platform**

Technologies: Python, SQL, Pandas, Scikit-learn, Power BI, Tableau

Built a data analytics and machine learning pipeline combining structured and semi-structured datasets for predictive analysis and automated data preparation.

Developed ETL, preprocessing, feature engineering, classification, and regression workflows using Python and SQL, reducing data preparation time by approximately 45%.

Created Power BI and Tableau dashboards to visualize analytical results and support data-driven decision-making.

CERTIFICATIONS

- Generative AI with Python and Hugging Face Transformers
- Computer Vision A–Z™: Learn OpenCV, SSD, YOLO, GANs and More
- AI Engineer Core Track: LLM Engineering, RAG, QLoRA, Agents
- Machine Learning, Data Science and Deep Learning with Python
- MLOps with MLflow, Docker, and Kubernetes for Deployment
- AI Engineer Production Track: Deploy LLMs & Agents at Scale
- AI Leader: Generative AI & Agentic AI for Leaders & Founders
- AI Builder: Create Agents, Voice Agents & Automations in n8n
- AI Engineer Agentic Track: The Complete Agent & MCP Course
- DevOps Fundamental CI/CD with AWS + Docker + Ansible + Jenkins
- Python for Data Science and Machine Learning Bootcamp
- TensorFlow Developer Certificate

PUBLICATION

[“Evaluation of dimensions of a vehicle using Velodyne and Blickfeld LiDAR”](#)

International Journal of Scientific and Engineering Research (IJSER), Volume 12, Issue 12, Dec 2021

DRIVING LICENSE

Class B